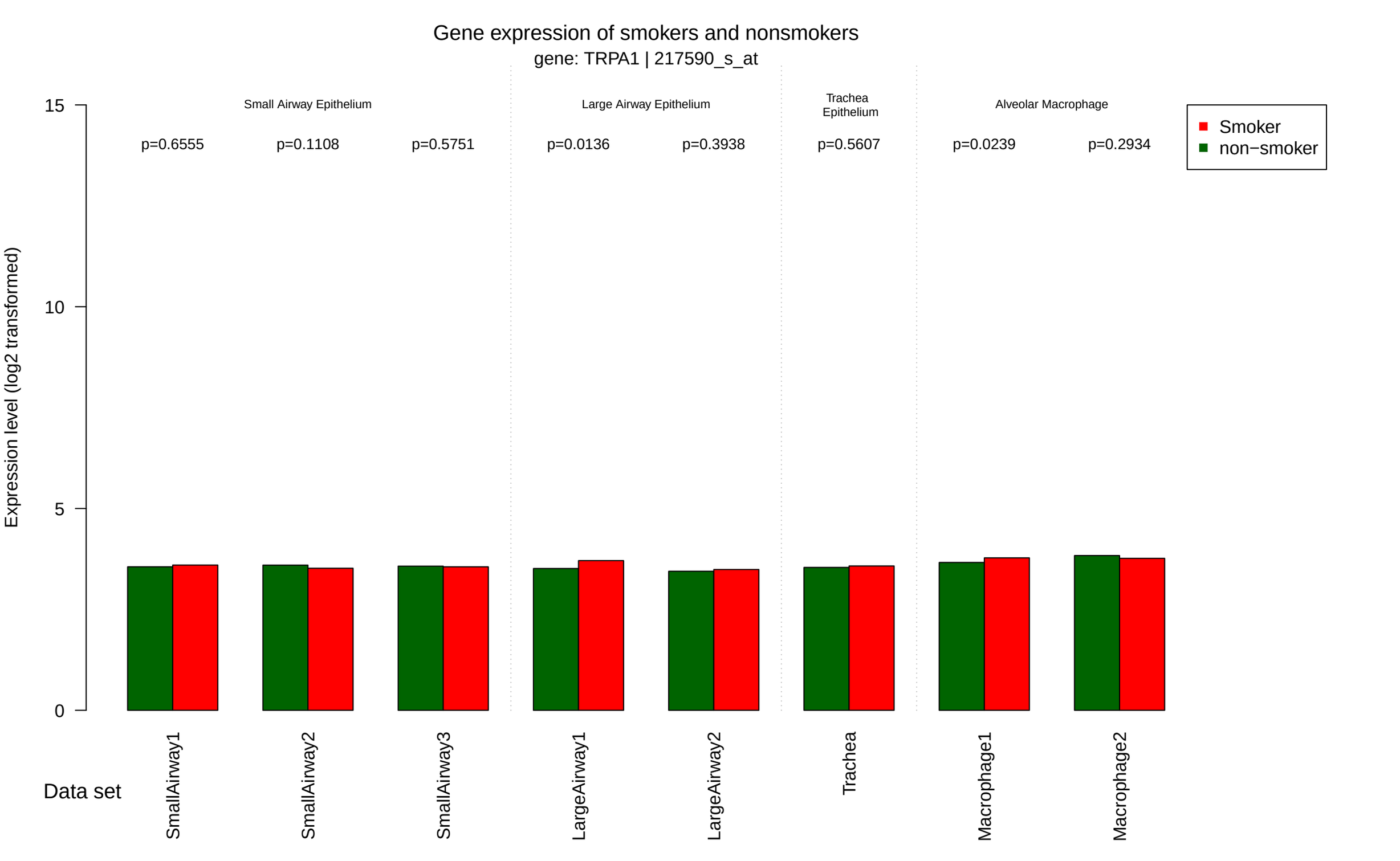
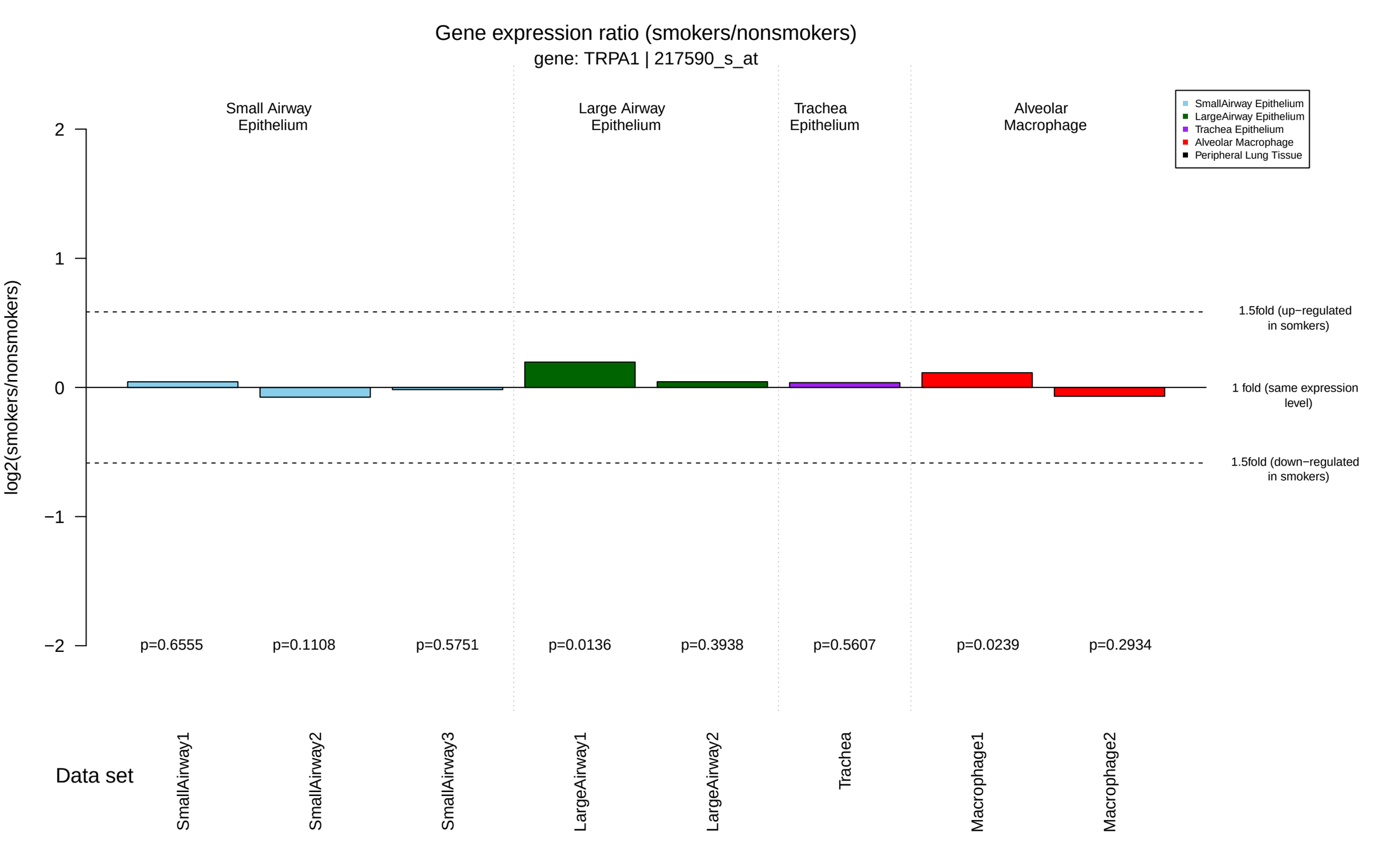
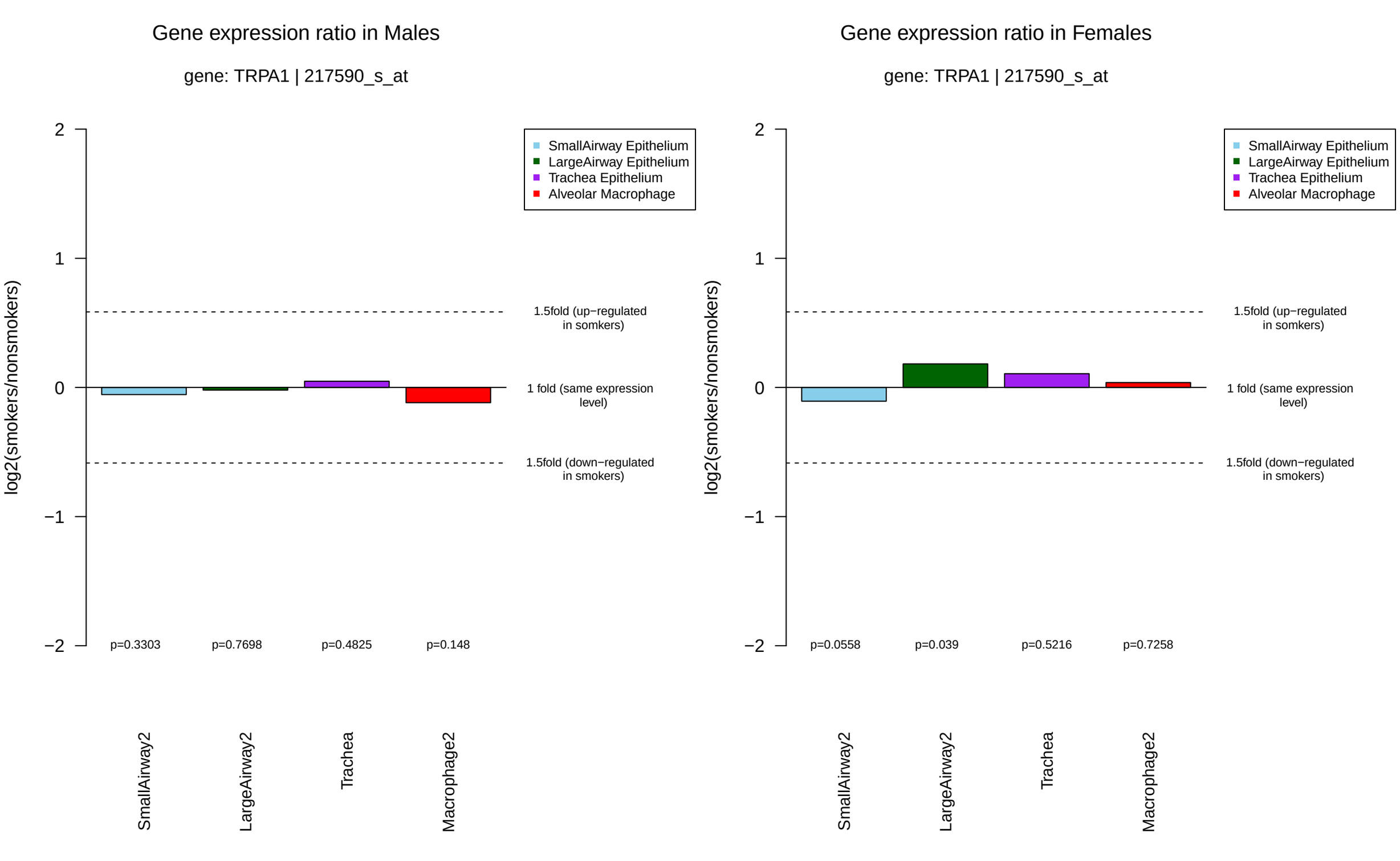
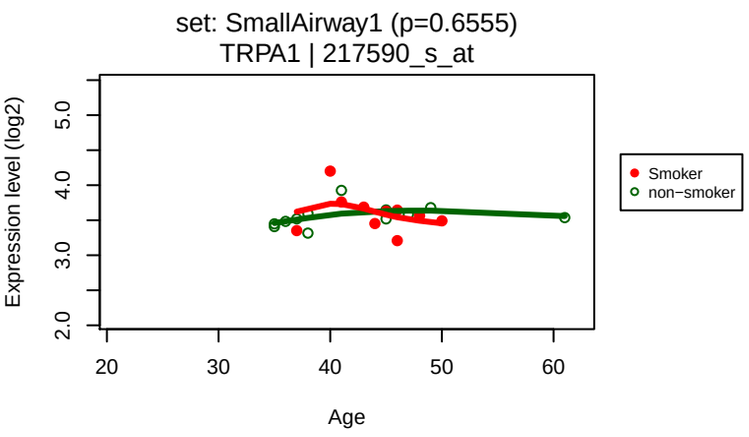
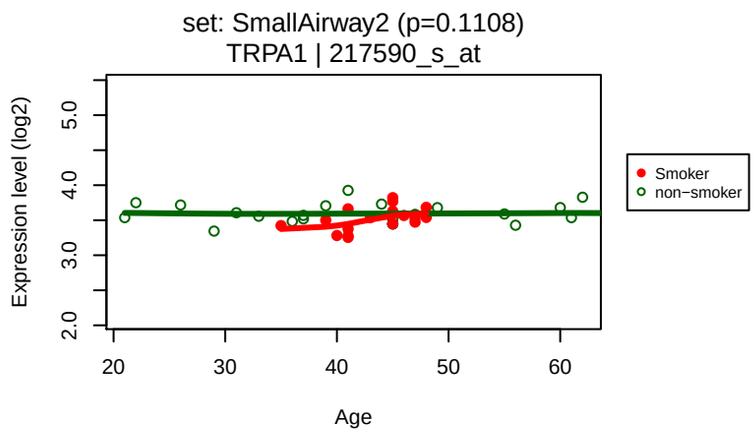
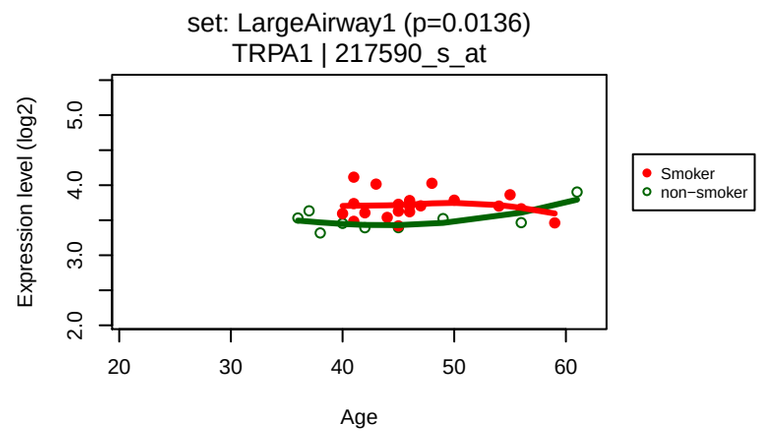

Gene expression of smokers and nonsmokers
gene: TRPA1 | 217590_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.6555
p=0.1108
p=0.5751
p=0.0136
p=0.3938
p=0.5607
p=0.0239
p=0.2934
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

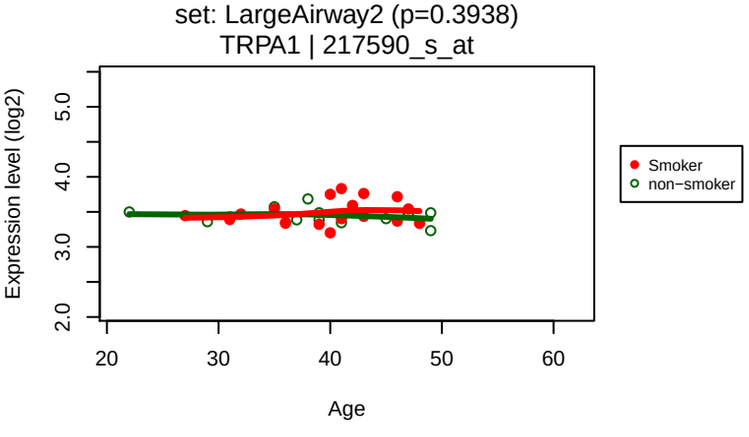

Gene expression ratio (smokers/nonsmokers)
gene: TRPA1 | 217590_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6555
p=0.1108
p=0.5751
p=0.0136
p=0.3938
p=0.5607
p=0.0239
p=0.2934
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

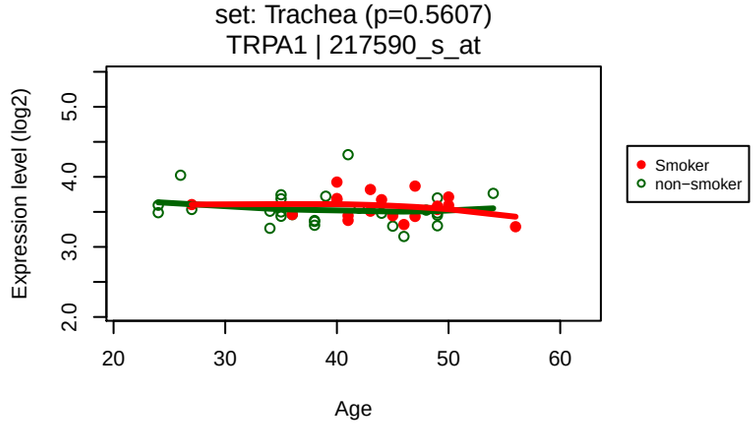

Gene expression ratio in Males
gene: TRPA1 | 217590_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3303
p=0.7698
p=0.4825
p=0.148
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: TRPA1 | 217590_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0558
p=0.039
p=0.5216
p=0.7258
SmallAirway2
LargeAirway2
Trachea
Macrophage2

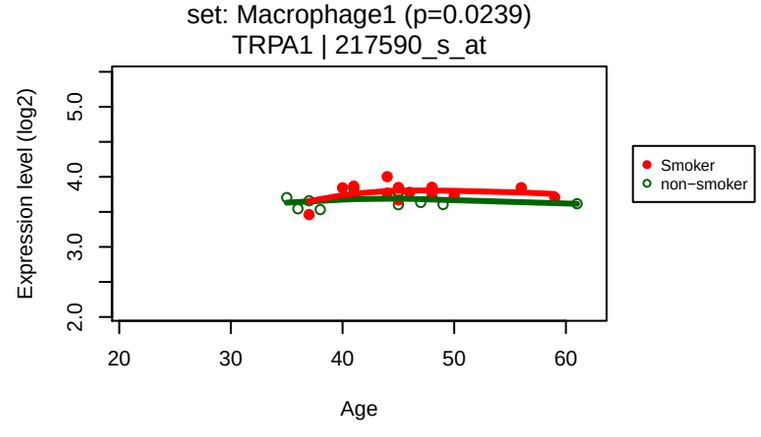

set: SmallAirway1 (p=0.6555)
TRPA1 | 217590_s_at
Expression level (log2)
Age
Smoker
non−smoker

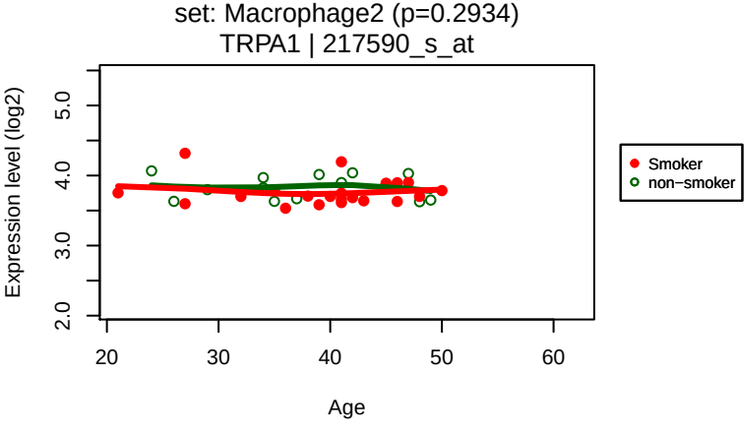

set: SmallAirway2 (p=0.1108)
TRPA1 | 217590_s_at
Expression level (log2)
Age
Smoker
non−smoker

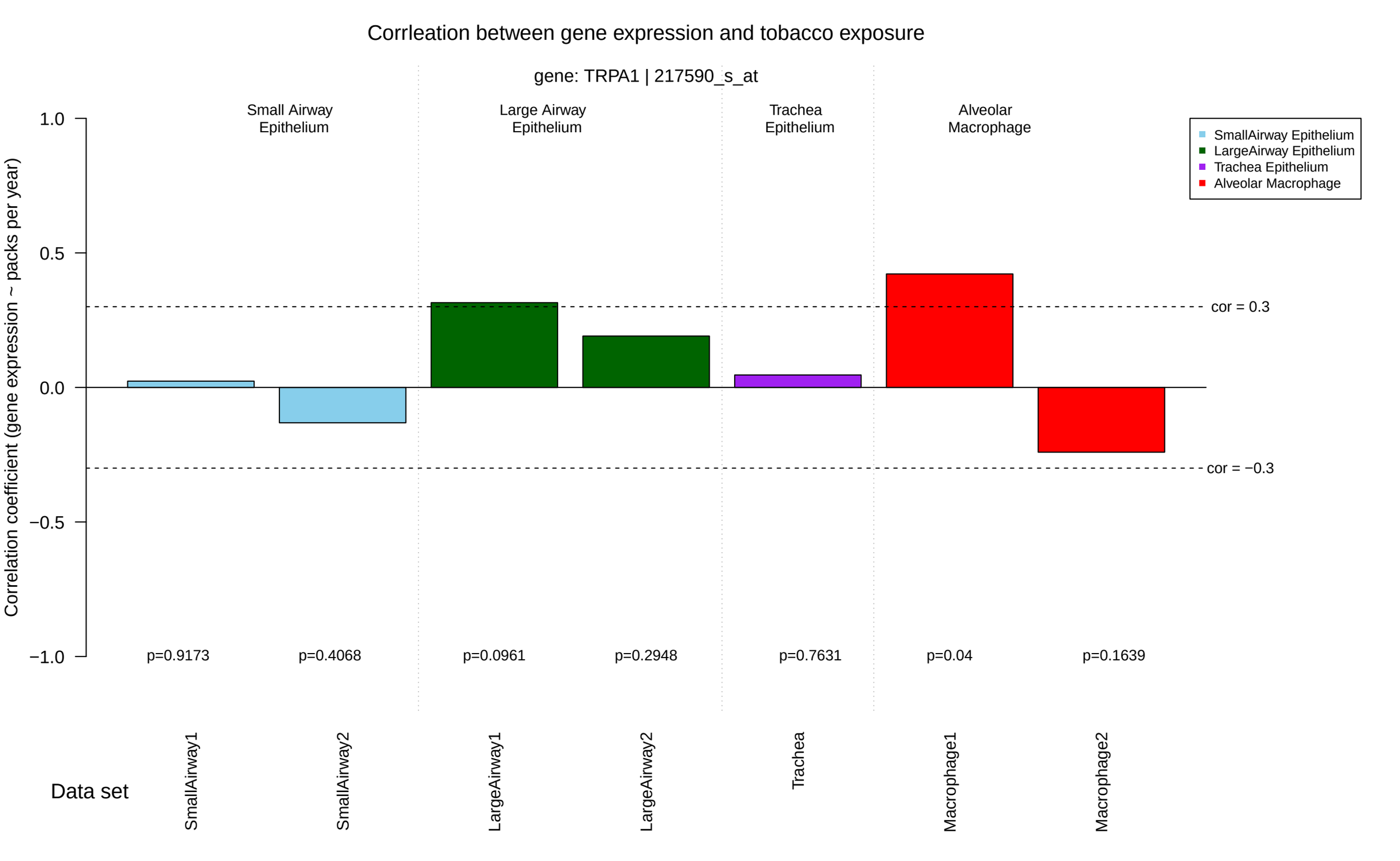

set: LargeAirway1 (p=0.0136)
TRPA1 | 217590_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.3938)
TRPA1 | 217590_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.5607)
TRPA1 | 217590_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0239)
TRPA1 | 217590_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2934)
TRPA1 | 217590_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: TRPA1 | 217590_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.9173
p=0.4068
p=0.0961
p=0.2948
p=0.7631
p=0.04
p=0.1639
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2